GNN-based SHM for H3 Rocket Payload Fairing

5-class defect detection + PINN regularisation + multitask centroid regression

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Outline

Problem setup

SHM problem: two complementary analyses

Structure: CFRP / Al-honeycomb sandwich fairing (H3 Type-S, 1/6 section) **Defects:** debonding / FOD / impact / delamination

(A) Static FEM analysis

- Abaqus static stress field per defect scenario
- **10,897 nodes × 34 features** (disp, stress, temp, layup, . . .)
- Graph: FEM connectivity
- Task: **5-class node classification**
  + defect centroid regression
- 700 simulations (560/140 split)
- Metric: OptF1 (threshold-optimal)

(B) Dynamic guided-wave (GW) analysis

- Abaqus explicit GW propagation (piezo excitation)
- **9 sensors × time series** (sampled at 4 MHz)
- Graph: sensor spatial network
- Task: **binary defect detection**
  + wave-phase PINN (Level 3)
- Dataset: 106 samples (100 healthy / 6 defect)
- Model: MIFNO-GW (2-D factorised FNO)

Static features enter the GNN as node attributes; GW time-series enter MIFNO-GW as a separate modality. Future: fused

Node features (34 dims)

Dims	Feature group	Physical meaning
0–2	Position (x, y, z)	Node coordinates
3–5	Normal (n_x, n_y, n_z)	Surface orientation
6–9	Curvatures (k_1, k_2, H, K)	Shell geometry
10–13	Displacement $(u_x, u_y, u_z, \|\|u\|\|)$	Structural response
14	Temperature	Thermal loading
15–18	Stress $(s_{11}, s_{22}, s_{12}, \sigma_{\text{Mises}})$	Failure indicators
19–23	Principal stress, thermal σ_M, log-strain	Derived quantities
24–30	Fibre direction, layup (0°/45°/–45°/90°)	Anisotropy
31–33	Circumferential angle, boundary/load flags	Topology

Methods

Three modelling additions

(A) Multi-task learning — defect centroid regression

Joint loss: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{reg}}\, \mathcal{L}_{\text{reg}}$

Shared backbone $\longrightarrow$ node classification head + graph-level MLP for (x, y, z) centroid.

GT centroid computed on-the-fly from defective nodes.

(B) Level-1 PINN — stress gradient regularisation

High predicted defect probability should co-locate with high $|\nabla \sigma_{\text{Mises}}|$.

$\mathcal{L}_{\text{stress}} = \frac{1}{N} \sum_i p_i^{(\text{def})} \, \exp(-|\Delta \sigma_i| / \sigma_0)$

(C) Level-2 PINN — static equilibrium residual $\nabla \cdot \sigma = 0$

Graph-edge finite differences on (s_{11}, s_{22}, s_{12}); penalises defect predictions at nodes with low equilibrium violation.

LGSTA — Local-Global Attention GNN

Best-performing architecture: LocalGlobalAttentionGNN (inspired by LGSTA-GNN, Buildings 2025)

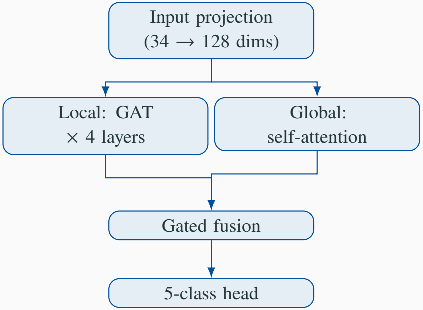
Local branch (GAT $\times$ 4 layers)

- Neighbourhood message-passing
- Captures local defect boundaries and stress concentrations

Global branch (Multi-head self-attention)

- Full $N \times N$ attention over all 10,897 nodes
- Captures global mode shapes and structural coupling

Fusion: gated combination of both representations



Memory: $O(N^2)$ global attention requires batch size = 1 on 24 GB GPU.

Results

Architecture & method comparison (interim, ep 40–100)

Model	Task	OptF1	AUC	deb	fod	imp	del	Reg. RMSE
LGSTA	NC+L1+L2	**0.862**	0.999	0.426	0.414	0.357	0.277	—
SAGE	NC+L2	0.845	0.999	0.298	0.381	0.404	0.002	—
LGSTA	MT+L1+L2	0.837	0.999	0.361	0.388	0.363	0.241	**0.8 mm**
GATv2	MT+L1+L2	0.829	0.999	0.342	0.315	0.359	0.000	0.7 mm
SAGE	MT+L1+L2	0.826	0.999	0.324	0.394	0.348	0.182	0.7 mm
GCN	MT+L1+L2	0.819	0.999	0.486	0.502	0.082	0.584	0.6 mm
GIN	MT+L1+L2	0.812	0.998	0.186	0.000	0.260	0.030	1.2 mm
Transolver	MT+L1+L2	0.010	0.493	—	—	—	—	—

- **NC** = node classification only **MT** = multitask (+centroid regression)
- **L1** = stress-gradient PINN **L2** = equilibrium-residual PINN
- Results at ep 40–100; training continues to 200 epochs

Key findings

1. LGSTA wins at 0.862 OptF1

Dual local-global attention captures both *local* stress concentrations and *global* structural coupling — especially important for detecting delamination patterns.

2. Multitask centroid regression converges to ~0.7–1.2 mm RMSE

With a single shared backbone, the model simultaneously classifies defect nodes *and* locates the defect centroid to sub-mm accuracy (normalised by 5,000 mm structure). Direct use in maintenance planning.

3. L2 PINN alone (SAGE, 0.845) is competitive with L1+L2

Static equilibrium residual $\nabla \cdot \sigma = 0$ provides effective physics regularisation even without the stress-gradient term.

Negative: Transolver collapses (AUC $\approx$ 0.49)

Transolver is designed for structured grids; it fails on the irregular FEM mesh topology even without physics losses.

Outlook

Next steps

- **Finish 200-epoch runs** — GCN, GIN, LGSTA, GATv2, MeshGNN convergence
- **Level-3 PINN** — wave phase consistency $\angle(\hat{u}_j \hat{u}_i^*) = \omega |dx| / c$ once guided-wave dataset expanded beyond 6 defect samples
- **Cross-structure evaluation** — train on fairing Section 1, test on Section 2 (OOD geometry)
- **Experimental data** — transfer to TSA / DIC measurements for sim-to-real gap analysis

Target for WCCM meeting (2026-06-23)

Final results table with all 8 architectures at 200 epochs + ablation: NC vs. MT, L1 vs. L2 vs. L1+L2.